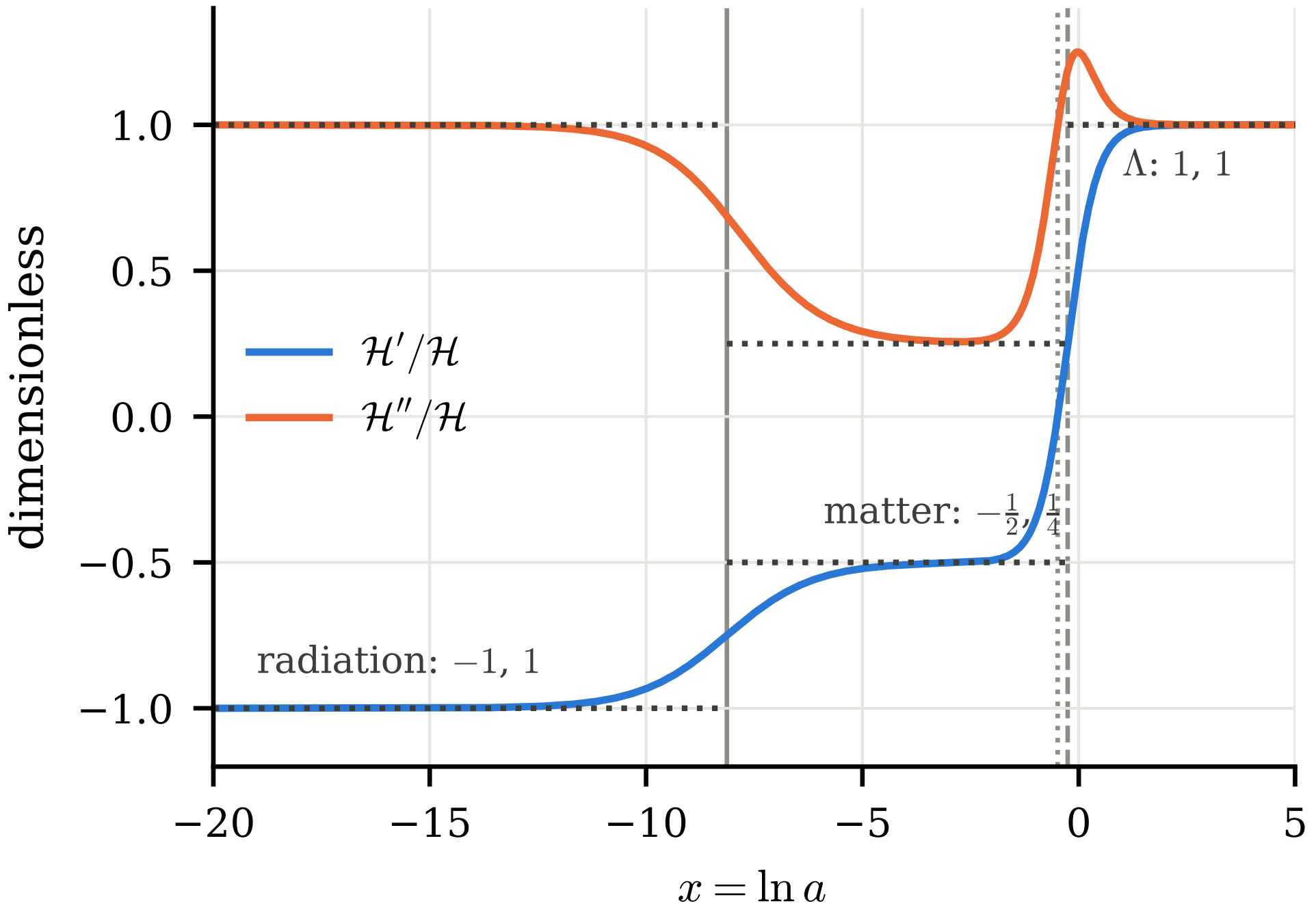
dimensionless
1.0
0.5
0.0
−0.5
−1.0
−20
−15
−10
−5
0
5
$x = \ln a$
$\mathcal{H}'/\mathcal{H}$
$\mathcal{H}''/\mathcal{H}$
radiation: −1, 1
matter: $-\frac{1}{2}, \frac{1}{4}$
Λ: 1, 1